React UI SDK Build Guide

This guide explains how to build the React UI SDK bundle for embedding in Angular (or other) applications.

Prerequisites

- Node.js >= 18.19.0
- npm or bun package manager
- Nx CLI (installed via npm)

Quick Build

From the workspace root (`/Users/rajarammohanty/Documents/POC/activepieces`):

```
# 1. Build the SDK bundle
npx nx run react-ui-sdk:bundle

# 2. Fix package.json (required for consumption outside the monorepo)
npx nx run react-ui-sdk:fix-bundle-package

# Or with skip cache (for fresh build)
npx nx run react-ui-sdk:bundle --skip-nx-cache
npx nx run react-ui-sdk:fix-bundle-package
```

Always run `fix-bundle-package` after `bundle`. It:

- Sets `main` and `exports` to `./index.js` (bundle output is at root, not in `dist/`).
- Replaces `@activepieces/shared` dependency `workspace:*` with `file:../../../packages/shared` so the bundle can be installed in non-workspace apps (e.g. Angular).

Output Location

After a successful build, the SDK bundle is located at:

```
dist/packages/react-ui-sdk-bundled/
├── assets/
│   ├── ActivepiecesCreateTodoGuide.png
│   ├── ActivepiecesTodo.png
│   ├── External_Channel_Todo.png
│   ├── ada-logo.png
│   └── fonts/           # Inter font woff2 (if added via project.json)
├── locales/             # i18n (ar, de, en, es, fr, ja, nl, pt, ru, zh, zh-TW)
│   ├── ar/translation.json
│   ├── en/translation.json
│   └── ...
├── *.woff2              # Webpack-emitted fonts (hashed names)
├── index.js             # Main SDK bundle (~12 MB, includes inlined styles)
├── LICENSE
├── README.md
└── package.json
```

Styles: Tailwind and react-ui global styles are imported in the bundle entry and inlined via style-loader. Host apps do not need to load a separate `styles.css` for the SDK.

Using the SDK from node_modules (Angular)

The recommended approach is to consume the bundle as an npm dependency and copy it into the app at build time.

1. In the Activepieces repo (first time and after SDK changes)

```
cd /path/to/activepieces
npx nx run react-ui-sdk:bundle
npx nx run react-ui-sdk:fix-bundle-package
```

2. In the Angular app

package.json – point to the bundled output:

```
{
  "dependencies": {
    "@activepieces/react-ui-sdk": "file:../activepieces/dist/packages/react-ui-sdk-bundled"
  }
}
```

Then run `bun install` or `npm install`.

angular.json – copy SDK files from node_modules into the build output (e.g. under `/sdk/`):

```
"assets": [
  { "glob": "index.js", "input": "node_modules/@activepieces/react-ui-sdk", "output": "sdk" },
  { "glob": "*.woff2", "input": "node_modules/@activepieces/react-ui-sdk", "output": "sdk" },
  { "glob": "locales/**/*", "input": "node_modules/@activepieces/react-ui-sdk", "output": "sdk" },
  { "glob": "assets/**/*", "input": "node_modules/@activepieces/react-ui-sdk", "output": "sdk" },
  { "glob": "LICENSE", "input": "node_modules/@activepieces/react-ui-sdk", "output": "sdk" },
  { "glob": "ada-logo.png", "input": "node_modules/@activepieces/react-ui-sdk/assets", "output": "." }
]
```

- `sdk/` holds the script, fonts, locales, and assets so they are served at e.g. `/sdk/index.js`, `/sdk/*.woff2`, `/sdk/assets/...`.
- `ada-logo.png` at root (`.`): the default platform theme returns `/ada-logo.png` for branding; copying it to the app root avoids a 404.

Before loading the SDK script, set the base URL so asset paths resolve correctly (webpack's runtime strips the last path segment from this URL to get `publicPath`):

```
(window as any).__AP_SDK_BASE_PATH__ = window.location.origin + '/sdk/index.js';
```

Then load the script: `<script src="/sdk/index.js">` (or equivalent). After it runs, use `window.__AP_SDK_MODULE__` (e.g. `mountReactComponent`, `Dashboard`, `FlowBuilder`, etc.).

API and WebSocket proxy – if the backend runs on another origin (e.g. `http://localhost:3000`), proxy `/api` and enable WebSocket:

```
// proxy.conf.json
{
  "/api": {
    "target": "http://localhost:3000",
    "secure": false,
    "changeOrigin": true,
    "pathRewrite": { "^/api": "" },
    "ws": true
  }
}
```

Set `apiUrl` to the app origin (e.g. `''` or `window.location.origin`) so that requests go to `/api/...` and are proxied to the backend.

Copying to Angular App (manual / bmp-fe-web)

If you prefer not to use node_modules (e.g. custom deployment or bmp-fe-web), copy the built files manually.

After building (and running `fix-bundle-package`), copy all SDK files into your project:

```
mkdir -p /path/to/bmp-fe-web/src/assets/sdk/{assets,locales}
cp dist/packages/react-ui-sdk-bundled/index.js /path/to/bmp-fe-web/src/assets/sdk/
cp dist/packages/react-ui-sdk-bundled/*.woff2 /path/to/bmp-fe-web/src/assets/sdk/ 2>/dev/null || true
cp -r dist/packages/react-ui-sdk-bundled/assets/* /path/to/bmp-fe-web/src/assets/sdk/assets/
cp -r dist/packages/react-ui-sdk-bundled/locales/* /path/to/bmp-fe-web/src/assets/sdk/locales/
```

For bmp-fe-web specifically:

```
mkdir -p /Users/rajarammohanty/Documents/ADA/bmp-fe-web/src/assets/sdk/{assets,locales} && \
cp dist/packages/react-ui-sdk-bundled/index.js /Users/rajarammohanty/Documents/ADA/bmp-fe-web/src/assets/sdk/ && \
cp dist/packages/react-ui-sdk-bundled/*.woff2 /Users/rajarammohanty/Documents/ADA/bmp-fe-web/src/assets/sdk/ 2>/dev/null || true &
cp -r dist/packages/react-ui-sdk-bundled/assets/* /Users/rajarammohanty/Documents/ADA/bmp-fe-web/src/assets/sdk/assets/ && \
cp -r dist/packages/react-ui-sdk-bundled/locales/* /Users/rajarammohanty/Documents/ADA/bmp-fe-web/src/assets/sdk/locales/
```

Note: Don't overwrite `bmp-overrides.css` or other custom styles you maintain in the host app.

Full Build Script (manual copy)

Example script to build and copy in one command:

```
#!/bin/bash
# build-sdk.sh

set -e

SDK_SRC="/Users/rajarammohanty/Documents/POC/activepieces"
SDK_DEST="/Users/rajarammohanty/Documents/ADA/bmp-fe-web/src/assets/sdk"

cd "$SDK_SRC"
npx nx run react-ui-sdk:bundle --skip-nx-cache
npx nx run react-ui-sdk:fix-bundle-package

mkdir -p "$SDK_DEST/assets" "$SDK_DEST/locales"
cp dist/packages/react-ui-sdk-bundled/index.js "$SDK_DEST/"
cp dist/packages/react-ui-sdk-bundled/*.woff2 "$SDK_DEST/" 2>/dev/null || true
cp -r dist/packages/react-ui-sdk-bundled/assets/* "$SDK_DEST/assets/"
cp -r dist/packages/react-ui-sdk-bundled/locales/* "$SDK_DEST/locales/"

echo "SDK built and copied successfully!"
ls -la "$SDK_DEST/"
```

Clean Build

If you encounter caching issues, clean and rebuild:

```
rm -rf dist/packages/react-ui-sdk-bundled
npx nx run react-ui-sdk:bundle --skip-nx-cache
npx nx run react-ui-sdk:fix-bundle-package
```

Build Targets

Target	Command	Output	Purpose
`build`	`npx nx run react-ui-sdk:build`	`dist/packages/react-ui-sdk`	TypeScript library (for npm)
`bundle`	`npx nx run react-ui-sdk:bundle`	`dist/packages/react-ui-sdk-bundled`	Webpack bundle (script tag)
`fix-bundle-package`	`npx nx run react-ui-sdk:fix-bundle-package`	(mutates bundle `package.json`)	Fix `main` and `@activepieces/shared` for external apps

Use `bundle` + `fix-bundle-package` for Angular (or any host) embedding via script tag.

Project Structure

```
packages/react-ui-sdk/
├── src/
│   ├── index.ts                # Main entry (npm package)
│   ├── index.browser.ts        # Browser entry (no Angular)
│   ├── index.browser.entry.ts  # Bundle entry (sets locales path, imports styles)
│   ├── react/                  # React component wrappers
│   ├── angular/                # Angular adapters
│   ├── providers/              # SDK providers
│   ├── types/                  # TypeScript types
│   ├── utils/                  # Utilities
│   ├── stubs/                  # Browser stubs for Node.js modules
│   └── scripts/
│       └── fix-bundle-package-json.js
├── project.json                # Nx project configuration
├── webpack.config.js           # Webpack bundle config
├── tsconfig.lib.json           # TypeScript config
└── package.json                # Package metadata
```

Adding Assets

To add new assets (images, fonts) to the SDK bundle:

1. Add the file under `packages/react-ui/src/assets/` (e.g. `img/custom/` or `assets/fonts/`).
2. Update `packages/react-ui-sdk/project.json` bundle target `assets` with the appropriate `glob` , `input` , and `output` .
3. Rebuild: `npx nx run react-ui-sdk:bundle` and `fix-bundle-package` if needed.

Customizing Styles

The bundle uses **style-loader**: Tailwind, theme variables, and Inter fonts are inlined and injected into the document when the script runs. There is no separate stylesheet file; changing how styles are loaded (e.g. extracting CSS) would require changing the SDK webpack config in this repo.

Default stylesheet: The inlined CSS includes Tailwind utilities, `@theme` and `:root` / `.dark` CSS variables (e.g. `--primary` , `--background` , `--foreground` , `--radius`), Inter `@font-face` , and component styles. Host apps can override these variables or load their own CSS **after** the SDK script to match their UI.

For full details (CSS variables to override, scoping, fonts, and examples), see **Default stylesheet and customizing the UI** in NPM_PACKAGE_USAGE.md.

Troubleshooting

Build fails with import errors

```
npm install   # or bun install
npx nx reset  # Clear Nx cache
```

Assets (fonts, images) 404 in host app

- Ensure the host sets `window.__AP_SDK_BASE_PATH__` to the **full script URL** (e.g. `https://example.com/sdk/index.js`) before loading the script, so webpack's `publicPath` becomes the correct base (e.g. `https://example.com/sdk/`).
- Ensure all needed globs are copied (e.g. `index.js`, `*.woff2`, `locales/**/*`, `assets/**/*`).

`/ada-logo.png` 404

The default platform theme returns `/ada-logo.png`. Either copy `ada-logo.png` from the bundle's `assets/` to the host app's root (e.g. via assets config) or change the platform branding to use another URL.

WebSocket connection failed

If the backend is on another origin, enable WebSocket proxying: in the proxy config for `/api`, set `"ws": true` and restart the dev server.

`workspace dependency "@activepieces/shared" not found` in host app

Run `fix-bundle-package` after `bundle` so the bundled `package.json` uses `file:../../../packages/shared` instead of `workspace:*` for `@activepieces/shared`.

TypeScript errors

```
npx nx run react-ui-sdk:build:skip-angular-check
```

Webpack bundle too large

The bundle is ~12 MB because it includes React, react-ui, and dependencies for standalone browser use. This is expected for script-tag loading.

Related Files

- Webpack config: `packages/react-ui-sdk/webpack.config.js`
- Project config: `packages/react-ui-sdk/project.json`
- Fix script: `packages/react-ui-sdk/scripts/fix-bundle-package-json.js`
- EE exclusion validator: `packages/react-ui-sdk/scripts/validate-ee-exclusion.ts`